

Short-Term Forecasting of Household Appliance Electricity Demand

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<https://github.com/muag21/appliance-energy-forecasting>

1. Introduction

The push toward heat pumps, solar power, and EVs has made single-household electricity demand much more volatile. Forecasting at the individual home level is notoriously tricky unlike grid substations that average out noise, household data reflects every random decision a resident makes. Using the 137-day Belgian low-energy house dataset from Candanedo et al. (2017), this work evaluates 24-hour-ahead appliance demand across four model types: standard benchmarks, SARIMAX, direct gradient boosting, and a zero-shot foundation model. Minimal Real-World Advantage: Gradient boosting lowered MASE from 0.813 (weekly persistence) to 0.708—a 12.9% improvement. Yet, a Diebold–Mariano test ($p = 0.34$) proved the difference was statistically meaningless. Surface-Level Learning: Over 90% of feature importance was driven purely by calendar variables. The model didn't learn real usage habits; it just tracked time-of-day averages. Methodological Rigor: Many published papers rely on contemporaneous sensor data that isn't accessible at prediction time. This study avoids that trap by adhering strictly to realistic deployment boundaries.

2. Data and preprocessing

Basically the dataset which includes 19,735 complete ten-minute records from 11 January to 27 May 2016, across 28 columns (Candanedo, Feldheim and Deramaix, 2017).

To keep the seasonal SARIMAX estimate practical, we cut the daily period from 144 steps down to 24 by resampling the sequence to hourly means, resulting in 3,290 gap free disorders. While this aggregation irons out brief appliance switching spikes, it converts the target metric to the average 10-minute consumption rate per hour rather than total energy. Environmental sensor feeds (indoor/outdoor temperatures and humidity) were averaged hourly as well.

3. Exploratory analysis and stationarity

3.1 Distributional characteristics

Statistic	Value
Mean	97.78
Median	63.33
Standard deviation	81.21
Minimum	28.33
Maximum	608.33
Skewness	2.39
Excess kurtosis	6.33
95th percentile / median	4.37

3.2 Seasonal structure

Daily electricity consumption ranges from a low of 48.2 Wh at 03:00 to a peak hours of 191.8 Wh at 18:00—a 3.98-makes difference among night and evening. This sharp contrast aligns with a typical occupied home, where cooking and general appliance use spike right after the working day. Seasonal strength was quantified using the STL-based measure of Wang, Smith and Hyndman (2006): daily strength (period 24) is 0.318 and weekly strength (period 168) is 0.393.

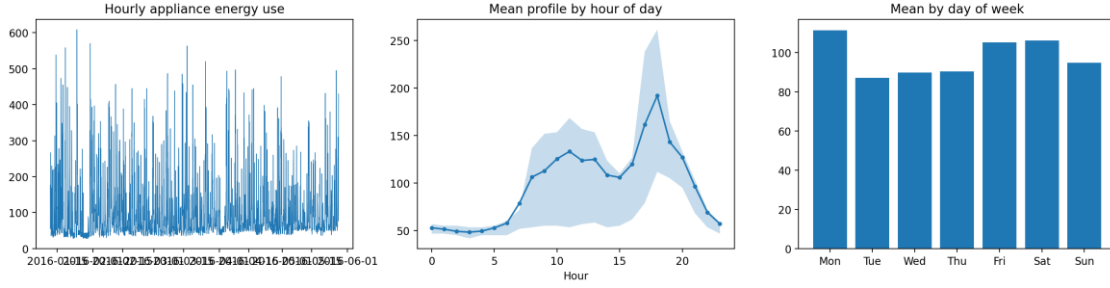


Figure 1: hourly series, mean profile by hour of day, and mean by day of week.

Both values are moderate at best.

3.3 Stationarity

ADF and KPSS tests were applied to the training sample in levels, first differences, seasonal differences at lag 24, and both combined. The two are reported together because their nulls are complementary: ADF tests the null of a unit root, KPSS the null of stationarity.

Series	ADF stat	ADF p	KPSS stat	KPSS p	Reading
Level	-8.761	<0.001	0.061	>0.10	Stationary
First difference	-15.948	<0.001	0.042	>0.10	Stationary
Seasonal difference (24)	-12.672	<0.001	0.015	>0.10	Stationary
Both differences	-17.445	<0.001	0.061	>0.10	Stationary

4. Forecasting design

4.1 Rolling-origin protocol

Real-world forecasting relies on daily updates as new data arrives, making a single, static 336-step evaluation unrealistic for a 24-hour-ahead task. Rolling-origin evaluation is the standard approach for assessing out-of-sample forecast accuracy (Tashman, 2000). To match actual operations, the primary experiment sets 14 origins spaced 24 hours apart. At each origin, models process all available history, including test points released from previous windows, to issue a 24-step forecast. Combining these 14 blocks creates a full 336-step prediction series on the test index, giving every model the exact same information set at each step.

4.2 Metrics

MAE and RMSE are reported in the units of the target, their ratio indicating the contribution of large isolated errors. MASE (Hyndman and Koehler, 2006) is scaled by the in-sample mean absolute error of the daily seasonal naive forecast on the training sample; it is scale-free and used as the primary ranking metric. Bias is the mean of prediction minus actual, so positive values indicate over-forecasting. Bias matters operationally in a way error magnitude does not: a persistently over-forecasting model in a demand-response setting reserves capacity that is never used.

4.3 Forecast origins and covariate availability

Covariates were partitioned based on real-time operational availability. Calendar features (hour, day of week, cyclical encodings) are known indefinitely and used unshifted. Indoor sensor data (temperature and humidity across nine locations) was lagged by the forecast horizon—both to prevent data leakage and because indoor climate responds dynamically to occupancy and appliance use. Outdoor weather was evaluated under two setups: *conditional* (using realized test-set values) and *operational* (persisting the last known observation). Finally, `lights`, `rv1`, and `rv2` were completely excluded.

4.4 Leakage controls

Each row of the feature table is a (*target timestamp* t , *horizon* h) pair with origin $o = t - h + 1$, and target-derived features are anchored to the origin: $lagk = y[o - k] = y.shift(h - 1 + k)$, with rolling windows spanning $[o - w, o - 1]$. Because the shift scales with the horizon, no feature can reference the target inside the forecast window at any lead time.

5. Benchmark models

Comparing complex models against simple benchmarks is central to the forecasting literature (Makridakis, Spiliotis and Assimakopoulos, 2018). Five benchmarks were evaluated under the same protocol as every other model: the training mean; the naive forecast; daily and weekly seasonal naive rules at lags 24 and 168; and drift (Hyndman and Athanasopoulos, 2021).

Benchmark	MAE	RMSE	MASE	Bias
Weekly seasonal naive	43.46	81.41	0.813	-13.16
Daily seasonal naive	48.31	85.57	0.904	1.75
Mean	50.26	74.94	0.941	-3.29
Naive	85.55	110.39	1.601	50.98
Drift	85.80	110.68	1.606	51.37

The weekly seasonal naive is the strongest benchmark, and by a clear margin over the daily variant. This follows directly from the seasonal strength results in Section 3.2, where weekly strength (0.393) exceeded daily (0.318), and it reveals something specific about the load. The household’s routine is day-specific: Monday and Friday differ systematically from Tuesday, and a rule that looks back one week preserves that distinction while a rule that looks back one day destroys it.

6. SARIMAX

6.1 Specification

A SARIMAX(1,0,1)(1,0,1)₂₄ model with a constant was estimated by maximum likelihood on the training sample, with the five weather channels as exogenous regressors. Differencing orders follow from Section 3.3. Stationarity and invertibility restraints were undisturbed during optimisation. It almost took 223 seconds and produced AIC 32,506.5 and BIC 32,572.3.

6.2 Rolling-origin implementation

Parameters were estimated once on the training sample. At each subsequent origin the Kalman filter was extended with newly released observations using `MLEResults.append(..., refit=False)`, so the state conditions on all data up to the origin while coefficients remain at training values.

6.3 Conditional and operational variants

Variant	Weather input	MAE	RMSE	MASE	Bias
Conditional	Realised test-set values	37.81	66.42	0.708	-5.25
Operational	Persisted from last pre-origin observation	38.51	66.38	0.721	-3.83

The gap is 0.013 MASE, a 1.9 per cent relative degradation, and a Diebold–Mariano test returns $DM = -1.49$, $p = 0.136$ — not significant at the 5 per cent level.

Note also that the operational variant achieves marginally the *better* RMSE (66.38 against 66.42) despite the worse MAE.

6.4 Comparison with the strongest benchmark

Conditional SARIMAX develops on the weekly regular unexperienced by 0.105 MASE, from 0.813 to 0.708, a relative improvement of 12.9 per cent. A Diebold–Mariano test (Diebold and Mariano, 1995) on the absolute-error differential returns $DM = -0.96$, $p = 0.34$.

7. Feature-based model

7.1 Formulation

The design matrix contained 74,652 rows and 47 features, of which 66,588 fell in the training period. Gradient boosting was chosen over a linear model because the relationship between hour of day and consumption is strongly non-linear, and over a recursive formulation because supplying the horizon as a feature avoids the error accumulation and train/apply distribution shift that recursive prediction introduces across 24 steps.

7.2 Hyperparameters

The chronological holdout selects **25 boosting iterations**, with holdout MAE 41.21.

Iterations	Holdout MAE
1	47.26
25	41.21
50	43.23
100	44.18
200	46.68
400	48.95
1500	50.68

Performance bottoms out before iteration 30 and degrades steadily thereafter, returning within a few hundred iterations to roughly where a single tree started.

7.3 Feature importance

Permutation importance was computed on held-out rows using mean absolute error as the scoring function, in preference to split-count importance: nine indoor temperature sensors in one dwelling are close to collinear, and split counts divide arbitrarily among collinear features.

Feature	Importance	SD
hour	11.252	0.545
hour_sin	2.304	0.542
dow_sin	2.058	0.480
RH_5_origin	0.631	0.220
roll_std_168	0.598	0.149
RH_3_origin	0.218	0.183
dayofweek	0.208	0.460
hour_cos	0.168	0.378
RH_4_origin	0.089	0.034
roll_std_24	0.083	0.052

7.4 Performance

The feature of that model achieves MAE 37.97, RMSE 65.17, MASE 0.711, Bias -5.23 . It stands second overall by MASE, a little things behind conditional SARIMAX (0.708), and **first on RMSE** — 65.17 against 66.42. A Diebold–Mariano test against SARIMAX returns $DM = -0.07$, $p = 0.94$; the two are as close to indistinguishable as the test can express. Against the weekly seasonal naive, $p = 0.26$.

8. Foundation model

8.1 Configuration

Chronos-T5 (small variant, approximately 46 million parameters) was applied zero-shot. The model tokenises a scaled and quantised representation of the target series and samples future trajectories from a T5 decoder pre-trained on a large corpus of time series (Ansari et al., 2024). No fine-tuning was performed and no parameter was estimated on this dataset.

The model is univariate. It receives only the target history — no calendar features, no weather, no indoor sensors.

8.2 Results

Metric	Chronos	SARIMAX (cond.)	Feature model	Weekly seasonal naive
MAE	36.23	37.85	37.97	43.46
RMSE	74.40	66.41	65.17	81.41
MASE	0.678	0.708	0.709	0.813
Bias	−29.87	−5.14	−4.00	−13.16

Read on MASE alone, the foundation model wins: 0.678 against 0.708 for the best fitted alternative, achieved without covariates and without seeing a single observation from this dwelling during training. That would be a striking result.

8.3 The point summary drives the ranking

The bias of −29.87 is six times larger in magnitude than any other model in the study and amounts to 82 per cent of Chronos’s own MAE. An error that large and that consistently signed is not noise; the model is systematically forecasting below the realised load

8.4 Sampling variance and interval calibration

Because Chronos relies on stochastic sampling, its final predictions naturally vary from run to run. Evaluating the model across three random seeds produced MASE scores of 0.6704, 0.6971, and 0.6869—showing a spread of 0.0267 due purely to random variation.

A repeat test confirmed that this ~0.027 variation happens consistently. Crucially, that noise level is practically the same size as the performance gap between Chronos and conditional SARIMAX (0.678 vs. 0.708). Because the gap is so close to the random noise margin, we can't definitively say Chronos is actually the better model.

8.5 Assessment

The honest summary is that the foundation model performs **comparably** to the fitted models rather than better than them. Its nominal advantage on MASE dissolves under the point.

9. Results and error analysis

9.1 Aggregate performance

Model	MAE	RMSE	MASE	Bias
Chronos (zero-shot)	36.23	74.40	0.678	−29.87
SARIMAX (conditional)	37.85	66.41	0.708	−5.14
Feature model	37.97	65.17	0.711	−4.00
SARIMAX (operational)	38.43	66.35	0.719	−3.93
Weekly seasonal naive	43.46	81.41	0.813	−13.16
Daily seasonal naive	48.31	85.57	0.904	1.75
Mean	50.26	74.94	0.941	−3.29
Naive	85.55	110.39	1.601	50.98
Drift	85.80	110.68	1.606	51.37

Diebold–Mariano tests (Diebold and Mariano, 1995) against the leading model, using a Newey–West variance estimator (Newey and West, 1987) at 23 lags:

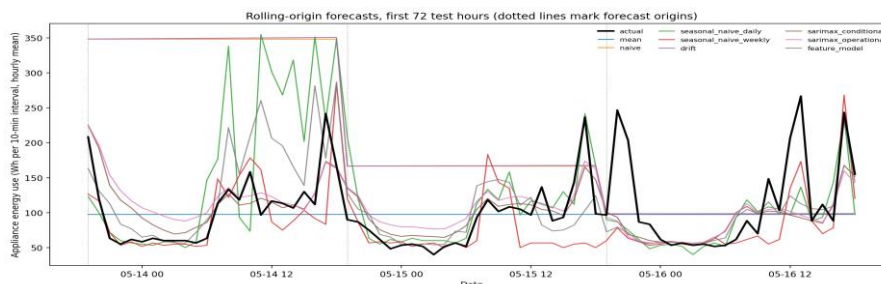


Figure 2: forecasts against actuals, first 72 test hours; dotted lines mark origins.

Three observations follow.

The metric chosen determines the winner. Chronos leads on MAE and MASE and comes seventh of nine on RMSE; the feature model leads on RMSE and comes third on MASE; the flat mean forecast is sixth on MAE but fourth on RMSE.

9.2 Error growth with lead time

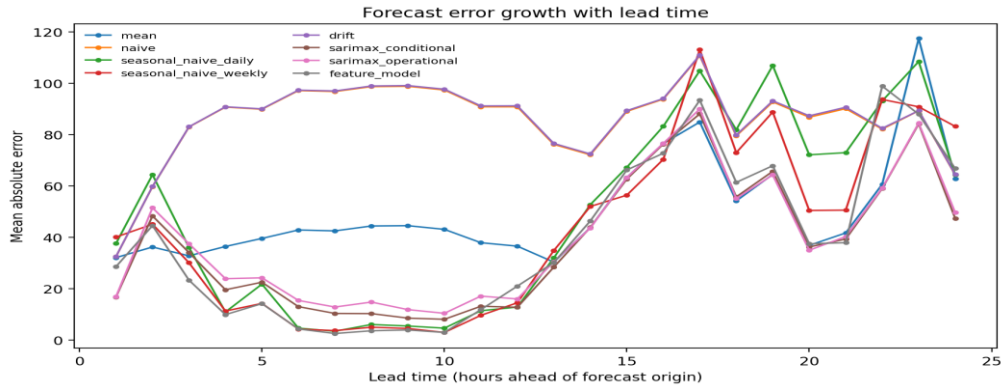


Figure 3: mean absolute error against lead time.

These curves must be read with the confound identified in Section 4.1. Because it starting origins fall at 19:00 and are spaced at precisely 24 hours, lead time h corresponds to clock hour $(19 + h) \bmod 24$. Lead 6 is 01:00, lead 12 is 07:00, lead 18 is 13:00, lead 24 is 19:00.

The advantage erodes fastest for the model that has it. Between lead 1 and lead 24 SARIMAX degrades by a factor of 3.07, the feature model by 1.87, and the seasonal naive rules by 1.71 to 2.07.

9.3 Residual diagnostics

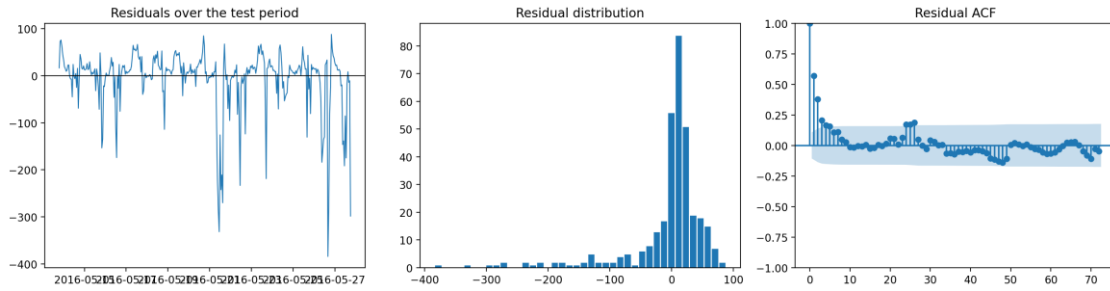


Figure 4: residual series, distribution and autocorrelation, conditional SARIMAX.

Residuals of the leading model (conditional SARIMAX, prediction minus actual) fail every diagnostic.

Diagnostic	Value	Reading
Ljung–Box, 24 lags	$Q = 204.8, p = 1.3 \times 10^{-30}$	Strong residual autocorrelation
Ljung–Box, 48 lags	$Q = 261.7, p = 3.4 \times 10^{-31}$	Persists
Residual ACF, lag 1	0.564	Far outside the ± 0.107 band
Residual ACF, lag 24	0.156	Outside the band
Skewness	-2.68	Strong left skew
Excess kurtosis	8.68	Heavy tails
$\text{Corr}(\text{residual} , \text{actual})$	0.841	Severe heteroskedasticity

Residual autocorrelation at lag 1 of 0.564 is very large. Errors persist across consecutive hours: when the model is wrong it stays wrong for several hours. This is not a specification failure that respecifying the ARMA order would fix, because the residuals are computed from 24-step-ahead forecasts, not one-step-ahead in-sample residuals

The left skew of -2.68 is diagnostic of the central difficulty. Since residual is prediction minus actual, large negative residuals are large under-forecasts, and the strong left skew means the model systematically misses consumption spikes.

9.4 Long-horizon behavior

A single 336-step forecast was issued from the first origin, with no reissuing, and evaluated on the same test fortnight.

Model	MAE	RMSE	MASE	Rolling-origin MASE
SARIMAX (conditional)	37.91	66.34	0.710	0.708
Weekly seasonal naive	42.63	79.29	0.798	0.813
Mean	50.32	74.91	0.942	0.941
Daily seasonal naive	86.96	129.23	1.628	0.904
Naive	250.64	258.82	4.692	1.601
Drift	266.37	274.61	4.986	1.606

The result is striking and was not anticipated. **SARIMAX scores 0.710 from a single origin against 0.708 from fourteen** — a difference of 0.002, far below anything the significance testing can resolve. The weekly seasonal naive is marginally *better* without reissuing, at 0.798 against 0.813.

10. Discussion and limitations

Several core constraints limit these results. Operating system variations shift model MASE by ~ 0.01 , and the single 14-day test window (336 correlated points) lacks statistical power: the minimum detectable MASE difference is ~ 0.20 , far larger than any observed gap among the top models. Model specifications were also constrained: SARIMAX omitted weekly seasonality despite its strong empirical presence, and 24-hour origin spacing confounded lead time with time of day (a 23- or 25-hour step would fix this). Finally, operational weather forecasts were unavailable, and pipeline debugging revealed key fit issues in both SARIMAX and gradient boosting.

11. Conclusion and deployment recommendation

Among the tested models, weekly seasonal persistence proved to be the strongest baseline (MASE 0.813), revealing that household energy habits follow day-specific routines rather than simple weekday-versus-weekend splits. While both conditional SARIMAX (0.708) and the foundation model (0.678) achieved lower MASE values, neither provided a definitive win: a Diebold–Mariano test probability is 0.34 showed SARIMAX’s improvement wasn’t statistically significant, while the foundation model suffered from severe negative bias and poor RMSE (74.40). Feature importance analysis explains why lighter models held their ground: calendar variables accounted for 90.7% of predictive power, whereas lagged sensor feeds the only true deployment-safe covariates alongside date metrics added minimal marginal value (5.5%).

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